

RAVEN: Erasing Invisible Watermarks via Novel View Synthesis

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Abstract

Invisible watermarking has become a critical mechanism for authenticating AI-generated image content, with major platforms deploying watermarking schemes at scale. However, evaluating the vulnerability of these schemes against sophisticated removal attacks remains essential to assess their reliability and guide robust design. In this work, we expose a fundamental vulnerability in invisible watermarks by reformulating watermark removal as a view synthesis problem. Our key insight is that generating a perceptually consistent alternative “view” of the same semantic content, akin to re-observing a scene from a shifted perspective, naturally removes the embedded watermark while preserving visual fidelity. This reveals a critical gap: watermarks robust to pixel-space and frequency-domain attacks remain vulnerable to semantic-preserving viewpoint transformations. We introduce a zero-shot diffusion-based framework that applies controlled geometric transformations in latent space, augmented with view-guided correspondence attention to maintain structural consistency during reconstruction. Operating on frozen pre-trained models without detector access or watermark knowledge, our method achieves state-of-the-art watermark suppression across 15 watermarking methods—outperforming 14 baseline attacks while maintaining superior perceptual quality across multiple datasets. Code: <https://github.com/fahadshamshad/raven-watermark-removal>.

1. Introduction

Invisible content watermarking has emerged as a key solution for tracking the provenance of AI-generated images [35, 42, 47]. These techniques embed imperceptible patterns that remain reliably detectable with a secret key even after common post-processing operations, a property typically referred to as *robustness*. Regulatory frameworks such as the EU AI Act [31] and the U.S. Executive Order on AI [3] now explicitly call for watermarking of AI-generated content to help prevent misuse and support authenticity verification. Major generative systems (e.g., SynthID from

Table 1. Comparison of watermark removal methods across three dimensions: (i) **Effectiveness**, i.e., how strongly the attack suppresses both pixel-level and semantic watermarks; (ii) **Quality**, preservation of semantic content and visual naturalness; and (iii) **Efficiency**, compute budget in training and per-image processing. Regen [49] is efficient but limited to weak watermarks; CtrlGen+ [25] requires multi-node training (8 GPUs); IRA [29] performs slow per-image optimization (≈ 40 min./image) and requires access to model parameters to be effective; and UnMarker [20] degrades visual fidelity. Our method achieves a favorable balance across all dimensions.

Approach	Effective	Quality	Efficiency
Regen _(NeurIPS’24) [49]	●●●●	●●●●	●●●●
CtrlGen+ _(ICLR’25) [25]	●●●●	●●●●	●●●●
IRA _(CVPR’25) [29]	●●●●	●●●●	●●●●
UnMarker _(S&P’25) [20]	●●●●	●●●●	●●●●
RAVEN (Ours)	●●●●	●●●●	●●●●

●●●● High ●●●● Moderate ●●●● Low

Google [15]) now incorporate watermarking into hundreds of millions of images. As the need for trusted AI-generated content continues to grow, evaluating the vulnerability of watermarking methods to removal attacks becomes critical to assess the usefulness of deployed systems and guide the development of more robust watermarking schemes [1, 11].

An ideal watermark removal attack must strike the right balance between perceptual fidelity and watermark suppression [17, 27, 48]. Perceptual fidelity refers to preserving the semantic content, fine-grained textures, and the absence of visible artifacts detectable by human observers or automated quality metrics. On the other hand, watermark suppression requires that the transformed image reliably evades detection, rendering the embedded signal statistically indiscernible to the watermark detector. A successful attack produces images that are perceptually identical to the original, while successfully defeating the detection mechanism [13].

Existing watermark removal attacks operate either in pixel space using signal-processing operations or in latent space through diffusion-based purification [40, 47]. However, both families struggle to jointly preserve perceptual fidelity and suppress watermark detectability (Tab. 1). Pixel-space methods, such as compression, filtering, and noise

injection, often fail against modern semantic watermarking schemes and can introduce visible artifacts [1]. Latent-space diffusion approaches can suppress stronger watermarks, but they typically rely on substantial noise injection, which disrupts scene structure and degrades visual realism [49]. In addition, many state-of-the-art methods incur high computational cost [25], requiring several minutes per image [29], or depend on privileged information such as access to the watermark decoder [28] or watermarked training data [33], which limits their practical applicability.

In this work, we expose a fundamental vulnerability in invisible watermarking by re-conceptualizing removal as a novel view synthesis problem [39]. Our key insight is that generating a perceptually consistent alternative “view” of the same semantic content produces a new image instance that preserves visual fidelity while statistically decorrelating from the embedded watermark signal. This exposes a critical gap in current robustness evaluations: even watermarks that can withstand pixel-space manipulation and latent-space purification remain vulnerable to semantic-preserving viewpoint transformations. Crucially, our approach does not require access to watermark detectors, knowledge of the watermarking scheme, or additional data or retraining, allowing deployment in practical settings.

To realize our view synthesis attack, we introduce a zero-shot diffusion-based framework that modulates latent representations to simulate novel viewpoints while preserving semantic content. Our method partially inverts the watermarked image into the latent space of the diffusion model, exposing the watermark-entangled representation while preserving the semantic structure of the scene. We then apply subtle latent-space viewpoint shifts that introduce geometric variations sufficient to disrupt watermark alignment without altering semantic image content. To maintain high perceptual fidelity during reconstruction, we replace conventional self-attention with a view-guided correspondence attention mechanism that enforces structural consistency between the original and synthesized views. Notably, our approach operates in a zero-shot manner using a frozen pre-trained image-to-image diffusion model and requires only the watermarked image as input, enabling deployment across diverse watermarking schemes without adaptation or retraining. Our contributions are as follows:

- **Novel Attack Vector via View Synthesis:** We expose a critical gap in watermark robustness evaluation by reframing removal as a view synthesis problem: synthesizing a perceptually consistent alternative view of the same scene preserves semantics while statistically decorrelating from the original watermark embedding.
- **Latent Viewpoint Modulation Framework:** We design a zero-shot diffusion-based pipeline that removes watermarks by applying controlled viewpoint shifts in latent space, producing perceptually consistent images without

requiring access to a watermark decoder or knowledge of the watermarking scheme.

- **View-Guided Correspondence Attention:** We introduce a correspondence-aware attention mechanism that maintains structural and appearance alignment between the original and synthesized views, enabling watermark suppression without altering fine-grained visual details.
- **Extensive Evaluation:** We demonstrate state-of-the-art watermark removal across multiple watermarking schemes and evaluate robustness under various attack conditions. Our approach attains superior watermark suppression with minimal perceptual degradation.

2. Related Work

Image Watermarking. Image watermarking enables provenance verification of AI-generated content by embedding imperceptible signals that can later be decoded for authentication [35, 42, 47]. Existing techniques can be grouped by when the watermark is inserted and where it resides in the representation. Post-hoc methods operate directly on generated images using signal processing or encoder-decoder architectures (e.g., DwtDCT [9], RivaGAN [45], StegaStamp [37]), offering broad applicability but often degrading under denoising, compression, or cropping. In contrast, in-generation watermarking modifies the generative model or its latent inputs, such as VAE-decoder tuning (StableSignature [14]), latent distribution shaping (Gaussian Shading [43]), or noise-pattern encoding (Tree-Ring [40]) to achieve stronger robustness by coupling the watermark with semantic structure. While these schemes are primarily evaluated for detection accuracy and robustness against distortions, their resilience to sophisticated removal attacks remains an active area of investigation [1].

Watermark Removal Attacks. Efforts to remove embedded watermarks can be characterized by their threat model assumptions and attack primitives. *Classic image processing methods* apply standard image transformations such as JPEG compression, geometric distortions, cropping, and additive noise [1]. While computationally efficient and requiring no model access, these approaches offer limited effectiveness against modern robust watermarking schemes [12]. *Regeneration attacks* leverage diffusion models [36, 49] or VAEs [28] to reconstruct images by injecting and removing noise, effectively suppressing pixel-level watermarks but often fail against semantic watermarks. Moreover, achieving removal of stronger watermarks requires substantial noise injection, which degrades structural consistency and visual realism, limiting practical applicability. A recently proposed *deep image prior* (DIP)-based method [23] removes watermarks without requiring generative models by exploiting the implicit regularization of untrained networks [30]. However, it is primarily effective against post-processing watermarks and strug-

gles with in-generation semantic schemes, whereas RAVEN achieves strong removal across both settings. Adversarial optimization-based methods treat the detector as a classifier and craft perturbations to induce detection failure [48], but typically rely on unrealistic capabilities—white-box access [28] or surrogate training using non-watermarked data [33, 41]. Existing approaches thus exhibit a persistent limitation: black-box methods struggle with semantic watermarks or quality preservation, while effective removal presumes impractical attacker capabilities.

Diffusion Models in Vision. Diffusion models have emerged as powerful generative frameworks for image synthesis and editing [19, 41], with recent advances enabling semantic-preserving transformations through controlled latent space manipulation. While diffusion-based NVS approaches excel at view synthesis and appearance editing, they generally require dedicated training on task-specific datasets with multi-view supervision [5, 44]. Recent zero-shot image-to-video frameworks demonstrate that pre-trained diffusion models can maintain semantic and structural consistency across transformed latent representations, even without retraining or paired data [21]. Motivated by this observation, we investigate whether similar latent viewpoint transformations can be leveraged for watermark removal, aiming to decorrelate embedded watermark signals while preserving perceptual identity. *To the best of our knowledge, we are the first to explore diffusion-based novel-view synthesis for invisible watermark removal.*

3. Threat Model

We formulate the watermark removal problem within a realistic threat model involving two key entities: a **provider** who operates a watermark-protected image generation service, and an **attacker** attempting to erase watermark evidence while retaining the semantic utility of the image.

Provider’s Goals and Capabilities: The provider watermarks the generated images to establish provenance and prevent misuse of generated content. They may adopt either post-hoc or in-generation watermarking schemes and possess full control over the deployed generator and watermark detector. Detection requires only a single image. We assume that the provider has sufficient computational resources and data to train a robust watermarking system.

Attacker’s Goals and Capabilities: The attacker aims to remove watermarks while preserving semantic content, producing alternative observations of the same scene rather than exact pixel reconstructions. The attacker operates under realistic constraints across four dimensions: (i) **Access:** no knowledge of model parameters, watermarking schemes, or detector internals; no access to detection APIs for iterative queries; (ii) **Data:** no paired clean-watermarked examples, no proprietary datasets for surrogate training, and access to only a single input image; (iii) **Compute:** lim-

ited resources for multi-GPU training or fine-tuning, limited to lightweight inference on consumer hardware; (iv) **Time:** per-image processing must complete within seconds, precluding minutes-long optimization. The attacker may use publicly available pretrained image-to-image diffusion models in a zero-shot manner, but lacks the compute or data to fine-tune or retrain them for removal or surrogate-detector training. This setting reflects practical scenarios where adversaries must succeed without privileged access, extensive datasets, or prolonged computation.

4. Proposed Approach

4.1. Background and Problem Statement

Image-to-Image Diffusion Models: We build our method on latent image-to-image diffusion models, instantiated with Stable Diffusion [32], which operate in the latent space of a pre-trained autoencoder $(\mathcal{E}, \mathcal{D})$. Let $x \in \mathcal{X}$ be an image sampled from the image space $\mathcal{X}$, $z_0 = \mathcal{E}(x) \in \mathcal{Z}$ be the corresponding latent from the latent space $\mathcal{Z}$ and $x' = \mathcal{D}(z_0) \approx x$. The forward process adds noise to latent z_0 via $q(z_t|z_0) = \mathcal{N}(z_t; \sqrt{\alpha_t}z_0, (1 - \alpha_t)\mathbf{I})$, while a learned denoiser $\epsilon_\theta(z_t, t)$ reverses this process to recover z_0 . Unlike text-to-image generation from random noise, image-to-image diffusion inverts a real image into noisy latent z_τ via DDIM inversion - optionally in a partial timestep $\tau < T$ to preserve the semantic structure - then applies transformations $\mathcal{T} : \mathcal{Z} \rightarrow \mathcal{Z}$ in latent space before reconstructing $x' = \mathcal{D}(\text{DDIM-Sample}(z'_\tau))$. This latent-space control enables image-to-image transformations that maintain semantic content while altering representational factors, making such models well suited for watermark removal attacks.

Watermarking System: We model a watermarking system as $\mathcal{W} = (\mathcal{U}_\zeta, \mathcal{V}_\eta)$, where encoder $\mathcal{U}_\zeta : \mathcal{X} \times \mathcal{K} \rightarrow \mathcal{X}$ embeds watermark key $\kappa \in \mathcal{K}$ into image x to produce watermarked image $x_w = \mathcal{U}_\zeta(x, \kappa)$, and detector $\mathcal{V}_\eta : \mathcal{X} \rightarrow \mathcal{K} \cup \{\perp\}$ extracts the embedded key $\hat{\kappa} = \mathcal{V}_\eta(x_w)$ or returns $\perp$ if no watermark is detected. Modern watermarks operate at two levels: *pixel-level* schemes embed signals in spatial or frequency domains (e.g., DwtDct [9], StegaStamp [37]), while *semantic-level* schemes couple signals with latent structure in generative models (e.g., Tree-Ring [40], StableSignature [14]) for stronger robustness. An effective watermark satisfies *imperceptibility* ($x_w \approx x$ perceptually) and *robustness* (invalidating κ requires noticeable degradation). In realistic deployments, both $\mathcal{U}_\zeta$ and $\mathcal{V}_\eta$ are proprietary, exposed only through black-box APIs.

Problem Formulation. Given a watermarked image $x_w = \mathcal{U}_\zeta(x, \kappa)$ returned by a provider’s generative API, our goal is to design a transformation $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{X}$ producing $\tilde{x} = \mathcal{A}(x_w)$ that removes the watermark (i.e., $\mathcal{V}_\eta(\tilde{x}) = \perp$ with high probability) while preserving the content (i.e., $\tilde{x} \approx x_w$). The attacker operates under strict no-box con-

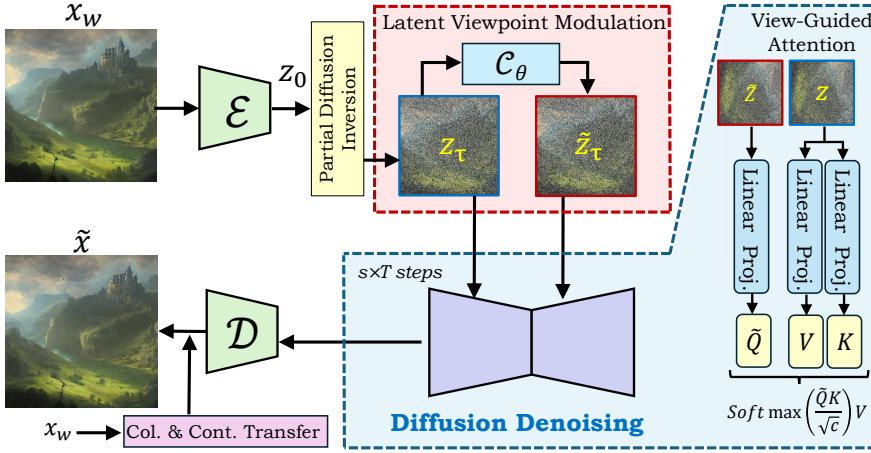


Figure 1. **Overview of RAVEN.** Given a watermarked image x_w , we encode it into latent space and perform partial diffusion inversion to obtain z_τ . Latent viewpoint modulation applies a camera motion C_θ to generate $\tilde{z}_\tau$. During denoising steps, view-guided correspondence attention computes queries $\tilde{Q}$ from the transformed latent while keys K and values V come from a reference latent denoised in parallel from z_τ . This cross-view mechanism preserves scene semantics while suppressing watermark-related cues. The decoder $\mathcal{D}$ reconstructs the output, followed by color and contrast correction to produce the final watermark-free result $\tilde{x}$.

straints (Sec. 3): no access to $(\mathcal{U}_\zeta, \mathcal{V}_\eta, \zeta, \eta, \kappa)$, no detector queries, no clean-watermark paired image supervision, but may use publicly available image-to-image diffusion models. Let $d : \mathcal{K} \times \mathcal{K} \rightarrow \mathbb{R}$ be an unknown distance metric in the key space $\mathcal{K}$. A successful attack must satisfy:

$$\text{(P1) Detection Evasion: } d(\mathcal{V}_\eta(\tilde{x}), \kappa) > \phi \quad (1)$$

$$\text{(P2) Semantic Preservation: } \mathcal{S}(\tilde{x}, x_w) \text{ is high} \quad (2)$$

$$\text{(P3) Visual Naturalness: } \mathcal{N}(\tilde{x}) \text{ is high} \quad (3)$$

where ϕ is the detection threshold, $\mathcal{S}$ denotes semantic similarity (e.g., CLIP-based), and $\mathcal{N}$ measures naturalness. We assume that the detector outputs $\perp$ when $d(\mathcal{V}_\eta(\cdot), \kappa) > \phi$, but the attacker does not know either d or ϕ . The removal problem thus requires jointly maximizing detection evasion (P1) while maintaining semantic consistency (P2) and naturalness (P3), objectives that are inherently coupled and often conflicting: stronger watermark suppression tends to degrade semantic or visual fidelity, while conservative edits may fail to evade detection. Striking the right balance between removal efficacy and content preservation is critical to effective removal of watermarks.

4.2. Our Method: RAVEN

Our goal is to remove invisible watermarks while preserving the semantic and perceptual integrity of the image. We approach this challenge by reframing watermark removal as a NVS problem, generating alternative observations of the same scene that remain semantically consistent but disrupt the spatial alignment and statistical coupling of the embedded watermark signal. This perspective departs from traditional signal-processing or advanced diffusion-based watermark removal methods, shifting the focus from erasing artifacts to reconstructing a perceptually consistent alternative view that, by virtue of its altered viewpoint, breaks the spatial and representational dependencies exploited by watermark encoders. One way to achieve this goal is to use recent

advanced NVS pipelines built on diffusion models trained for 3D-aware generation or video synthesis [5, 44]. However, such frameworks rely on extensive multi-view training data and high computational resources, making them impractical for watermark removal where paired supervision and clean references are unavailable.

Overview of Proposed Solution: To overcome these limitations, we propose RAVEN, a zero-shot diffusion-based framework designed to remove watermarks without requiring access to a watermark detector, embedding keys, or paired clean-watermarked training data. Our key insight is that pre-trained image-to-image diffusion models already encode rich geometric and semantic priors sufficient for view synthesis—they simply require lightweight modifications to maintain consistency across synthesized views. As illustrated in Fig. 1, RAVEN operates in three stages: (i) **partial diffusion inversion**, where we encode the watermarked image into a noisy latent representation, preserving semantic structure while exposing the watermark-entangled representation; (ii) **latent viewpoint modulation**, where we apply controlled geometric transformations to simulate viewpoint shifts that disrupt watermark alignment; and (iii) **view-guided correspondence attention**, which replaces standard self-attention with cross-frame attention mechanisms during the denoising process to preserve structural consistency. The following sections describe these stages in detail.

4.2.1. Partial Diffusion Inversion

Given a watermarked image x_w , we first map it into the diffusion model’s latent space using the encoder $\mathcal{E}(\cdot)$, producing a latent code $z = \mathcal{E}(x_w)$. We apply partial diffusion inversion to introduce controlled noise into the latent while keeping its overall structure intact. To achieve this, we add noise corresponding to an intermediate diffusion timestep $\tau = \lfloor s \cdot T \rfloor$, where T is the total number of diffusion steps and $s \in [0, 1]$ controls the noise level. The forward diffusion update is:

$$z_\tau = \sqrt{\bar{\alpha}_\tau} z + \sqrt{1 - \bar{\alpha}_\tau} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (4)$$

where $\bar{\alpha}_\tau = \prod_{t=1}^\tau \alpha_t$ denotes the cumulative noise schedule. The parameter s controls the trade-off between noise strength and semantic preservation: small values of s retain finer details, whereas large values introduce more stochasticity at the cost of potential semantic drift. The resulting latent z_τ maintains the overall scene structure while injecting the controlled randomness required for the subsequent latent viewpoint-modulation stage.

4.2.2. Latent Viewpoint Modulation

Given the partially inverted latent z_τ , our goal is to introduce a small, semantically consistent viewpoint change directly in latent space. Such a shift acts like a subtle camera motion: it displaces scene structures coherently while preserving the overall layout. We model this operation using a spatial warping function $\mathcal{C}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, which specifies the sampling location for each latent feature. The modulated latent $\tilde{z}_\tau$ is obtained as:

$$\tilde{z}_\tau[i, j] = z_\tau[\mathcal{C}_\theta(i, j)]. \quad (5)$$

Here, $\tilde{z}_\tau$ is a geometrically transformed version of z_τ , where each spatial location (i, j) in the output latent fetches its content from a shifted location determined by $\mathcal{C}_\theta(i, j)$. This warping operation induces a coherent displacement of scene structures in the latent space—analogueous to observing the same scene from a slightly shifted camera position, while preserving the global semantics encoded in z_τ .

Conventional NVS pipelines handle large viewpoint changes and therefore rely on multi-view training data, depth estimation, and heavy 3D consistency modeling [5, 44]. Such complexity is unnecessary for our goal. We do not seek to reconstruct unseen geometry, but only to introduce a small, coherent perturbation in viewpoint that slightly alters spatial correspondence while preserving semantic content. For example, a simple and effective choice is a *global camera translation*, $\mathcal{C}_\theta(i, j) = (i + \Delta_x, j + \Delta_y)$, where (Δ_x, Δ_y) determines the direction and magnitude of the apparent camera motion. This transformation induces a subtle yet spatially coherent shift in scene structure while maintaining semantic stability. Although more general motions are possible, small global translations strike the best balance between simplicity and consistency in our setting.

4.2.3. View-Guided Correspondence Attention

While latent viewpoint modulation introduces the spatial perturbation needed to disrupt watermark alignment, naively denoising the transformed latent $\tilde{z}_\tau$ is insufficient. Without guidance, the diffusion model treats $\tilde{z}_\tau$ as an independent image, causing appearance drift, color shifts, and loss of fine details. To preserve perceptual correspondence

with the original scene, we modify the attention mechanism within the diffusion UNet. In standard diffusion models $\epsilon_\theta(z_t, t)$, self-attention operates only on the current noisy latent $\text{SelfAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$, where $Q = W_Q z_t$, $K = W_K z_t$, and $V = W_V z_t$. This restricts all interactions to a single frame and provides no mechanism to anchor denoising to the original content after a viewpoint transformation.

We introduce view-guided attention that couples the transformed latent $\tilde{z}_t$ with a reference latent z_t^{ref} denoised in parallel from the untransformed inverted latent z_τ . At each timestep t , queries are computed from the transformed latent while keys and values come from the reference:

$$\text{ViewAttn}(Q, K, V) = \text{softmax}\left(\frac{(W_Q \tilde{z}_t)(W_K z_t^{\text{ref}})^\top}{\sqrt{d}}\right) W_V z_t^{\text{ref}} \quad (6)$$

This mechanism forces each position in $\tilde{z}_t$ to attend to semantically corresponding features in z_t^{ref} , preserving appearance, texture, and object identity throughout denoising. Critically, the attention operates in learned feature space rather than pixel coordinates—each query matches to regions in the reference based on semantic similarity, naturally accommodating the spatial misalignment introduced by viewpoint modulation. This allows the diffusion model to reconstruct a perceptually coherent alternative view while the watermark, which depends on precise pixel-level spatial correlation, remains disrupted. The view-guided correspondence enforced at each denoising timestep prevents semantic drift, ensuring geometric perturbations selectively remove the watermark without degrading scene fidelity.

4.2.4. Color and Contrast Transfer

While view-guided correspondence attention preserves structural and semantic fidelity, minor color shifts or contrast inconsistencies may still arise due to stochastic denoising. We found that a simple color and contrast transfer in the CIELAB space effectively restores perceptual quality without reintroducing the watermark. Let $x_{\text{opt}} = \mathcal{D}(\tilde{z}_0)$ denote the denoised image with CIELAB components $\{L_{\text{opt}}, a_{\text{opt}}, b_{\text{opt}}\}$, and let $\{L_w, a_w, b_w\}$ denote the corresponding components of the watermarked image x_w . For **color transfer**, we retain the luminance from the optimized image and adopt the chrominance from the watermarked one: $x_c = \mathcal{F}_{\text{RGB}}(L_{\text{opt}}, a_w, b_w)$, where $\mathcal{F}_{\text{RGB}}$ converts from CIELAB to RGB space. For **contrast transfer**, we align the luminance statistics between x_c and x_w as $L_{\text{final}} = \frac{\sigma_w}{\sigma_c}(L_{\text{opt}} - \mu_c) + \mu_w$, where μ_c, σ_c and μ_w, σ_w are the mean and standard deviation of the luminance channels of x_c and x_w , respectively. The final image is reconstructed as $\tilde{x} = \mathcal{F}_{\text{RGB}}(L_{\text{final}}, a_w, b_w)$. These simple steps effectively correct color artifacts and enhance local contrast, complementing the diffusion-based stages.

Table 2. Verification performance of different watermarking methods under various attacks. **TPR@1%FPR** is reported for in-generation semantic watermarking methods (Tree-Ring to ROBIN), where lower values indicate better attack performance. **Bit Accuracy** is reported for post-hoc bitstream-based methods (DwtDct to VINE), where values near 0.5 indicate successful watermark randomization. RAVEN achieves the lowest detection rates across both categories, demonstrating superior removal efficacy while maintaining visual quality.

Attack	Tree-Ring [40]	Zodiac [46]	HSTR [22]	RingID [8]	HSQR [22]	ROBIN [18]	Avg.	DwtDct [9]	DwtDctSvd [9]	RivaGAN [45]	Stable Sign. [14]	Gauss. Shad. [43]	TrustMark [4]	Stega St. [37]	VINE [26]	Avg.
MS-COCO [24]																
Clean	0.957	0.998	1.000	1.000	1.000	1.000	0.993	0.863	1.000	0.999	0.995	1.000	0.952	1.000	1.000	0.968
Bright.	0.463	0.843	0.899	0.988	0.991	0.985	0.862	0.572	0.555	0.862	0.894	0.962	0.918	0.992	0.998	0.839
Cont.	0.900	0.998	1.000	1.000	1.000	0.994	0.982	0.522	0.473	0.986	0.978	1.000	0.927	1.000	0.998	0.876
JPEG	0.548	0.973	0.994	1.000	1.000	0.986	0.917	0.516	0.602	0.821	0.806	0.992	0.919	1.000	1.000	0.851
Blur	0.934	0.998	1.000	1.000	1.000	0.991	0.987	0.677	1.000	0.998	0.911	1.000	0.942	0.990	0.973	0.944
Noise	0.412	0.880	0.806	0.987	0.983	0.840	0.818	0.859	1.000	0.969	0.721	0.997	0.772	0.971	0.979	0.878
BM3D	0.815	0.997	0.999	1.000	1.000	0.863	0.946	0.532	0.784	0.934	0.838	0.999	0.886	0.997	1.000	0.886
Center Crop	0.509	0.989	1.000	1.000	1.000	0.884	0.897	0.729	0.744	0.991	0.987	0.998	0.890	0.924	0.953	0.890
Random Crop	0.734	0.995	1.000	1.000	1.000	0.790	0.920	0.810	0.861	0.995	0.991	1.000	0.724	0.887	0.915	0.898
VAE-B [2]	0.509	0.944	0.973	0.992	0.992	0.817	0.871	0.523	0.648	0.570	0.717	0.982	0.834	0.883	0.950	0.788
VAE-C [6]	0.536	0.958	0.982	1.000	1.000	0.820	0.883	0.521	0.596	0.552	0.715	0.987	0.847	0.870	0.915	0.778
Regen. [49]	0.543	0.972	0.997	1.000	1.000	0.873	0.898	0.519	0.644	0.608	0.478	0.999	0.790	0.854	0.881	0.752
Rinse	0.461	0.905	0.982	0.997	0.999	0.793	0.856	0.508	0.561	0.527	0.494	0.999	0.713	0.809	0.831	0.716
CtrlGen+ [25]	0.090	0.314	0.776	1.000	1.000	0.328	0.585	0.532	0.517	0.481	0.564	1.000	0.683	0.557	0.887	0.640
UnMarker [20]	0.032	0.092	0.031	0.257	0.018	0.039	0.078	0.483	0.528	0.553	0.511	0.592	0.547	0.636	0.617	0.573
RAVEN (Ours)	0.020	0.067	0.025	0.018	0.015	0.012	0.026	0.517	0.472	0.499	0.519	0.540	0.476	0.575	0.588	0.533
SD-Prompts [34]																
Clean	0.944	0.998	1.000	1.000	1.000	1.000	0.990	0.819	1.000	0.991	0.994	1.000	0.970	1.000	1.000	0.972
Bright.	0.471	0.748	0.742	0.972	0.955	0.999	0.815	0.557	0.537	0.823	0.899	0.949	0.922	0.992	1.000	0.835
Cont.	0.894	0.999	1.000	1.000	1.000	1.000	0.982	0.516	0.459	0.963	0.967	0.999	0.928	0.988	0.983	0.850
JPEG	0.466	0.979	0.990	1.000	0.999	0.989	0.904	0.506	0.610	0.810	0.769	0.990	0.936	1.000	1.000	0.828
Blur	0.912	0.999	1.000	1.000	1.000	0.981	0.982	0.685	0.999	0.988	0.888	0.998	0.910	0.994	0.980	0.930
Noise	0.423	0.903	0.850	0.988	0.992	1.000	0.859	0.822	0.998	0.961	0.742	0.992	0.819	0.960	0.948	0.905
BM3D	0.802	1.000	1.000	1.000	1.000	0.876	0.946	0.530	0.859	0.915	0.809	1.000	0.864	1.000	0.971	0.869
Center Crop	0.469	0.994	1.000	1.000	1.000	0.860	0.887	0.723	0.743	0.980	0.983	1.000	0.872	0.893	0.920	0.889
Random Crop	0.749	0.996	1.000	1.000	1.000	0.799	0.924	0.794	0.860	0.983	0.990	1.000	0.749	0.891	0.912	0.897
VAE-B [2]	0.509	0.940	0.983	0.996	0.996	0.849	0.879	0.513	0.659	0.572	0.677	0.976	0.820	0.853	0.862	0.742
VAE-C [6]	0.514	0.975	0.987	1.000	0.999	0.881	0.893	0.512	0.620	0.535	0.671	0.994	0.829	0.881	0.900	0.743
Regen. [49]	0.543	0.958	0.999	1.000	1.000	0.857	0.893	0.509	0.623	0.567	0.493	0.999	0.803	0.852	0.890	0.717
Rinse	0.498	0.872	0.967	0.983	0.991	0.783	0.849	0.493	0.547	0.528	0.511	0.951	0.751	0.839	0.872	0.687
CtrlGen+ [25]	0.076	0.283	0.768	1.000	1.000	0.127	0.542	0.494	0.471	0.548	0.591	1.000	0.655	0.562	0.813	0.642
UnMarker [20]	0.028	0.084	0.036	0.284	0.024	0.043	0.083	0.485	0.507	0.548	0.511	0.579	0.523	0.658	0.604	0.552
RAVEN (Ours)	0.025	0.062	0.038	0.024	0.020	0.018	0.031	0.471	0.504	0.477	0.523	0.562	0.480	0.587	0.600	0.526

5. Experiments

5.1. Experiment Settings

Model Setup and Datasets. We evaluate RAVEN on three datasets: 5,000 captions from MS-COCO [24], 1,001 prompts from DiffusionDB [38], and 8,192 prompts from SD-Prompts [34]. Images are generated at 512×512 resolution using Stable Diffusion v2-1 [32] with CFG scale 7.5 and 50 DDIM steps. Watermark detection uses 1,000 image pairs to measure TPR@1%FPR.

Watermark Methods. We evaluate RAVEN on 15 watermarking methods using official implementations, spanning *post-hoc* and *in-generation* schemes: DwtDct [9], DwtDctSvd [9], RivaGAN [45], StegaStamp [37], StableSignature [14], Tree-Ring [40], Zodiac [46], HSTR [22], RingID [8], HSQR [22], Gauss. Shading [43], ROBIN [18], TrustMark [4], and VINE [26]. Most methods are designed with robustness as a core objective, providing a rigorous benchmark for RAVEN’s evaluation.

Image Watermark Removal Baselines. We evaluate

RAVEN against 14 watermark removal attacks across three categories. (i) *Classical Image Processing*: brightness adjustment, contrast modification, JPEG compression, Gaussian blur, additive Gaussian noise, and BM3D denoising [10]. (ii) *Regeneration Attacks*: Regen [49] reconstructs images via diffusion-based noising and denoising; Rinse applies Regen iteratively; VAE-B [2] and VAE-C [6] use VAE-based compression at quality level 3. (iii) *Advanced Methods*: CtrlGen+ [25] performs controllable regeneration from clean-noise initialization, while UnMarker [20] is a universal optimization-driven attack effective across diverse watermarking schemes. We exclude methods requiring surrogate models, such as IRA [29] or adversarial attacks [1], as their performance depends on surrogate quality whereas RAVEN operates in a black-box setting.

Implementation Details. We use a frozen Image-to-Image Stable Diffusion Refiner with CFG scale 2.5, 50-step DDIM inversion, and image-to-image strength 0.15. Since the original prompt is unknown, we use empty text prompts for inversion and reconstruction. Latent viewpoint modu-

Metric	Signal Processing					Regeneration					Advanced		Ours
	Bright.	Blur	Noise	R.C.	C.C.	VAE-B	VAE-C	Regen.	Rinse2x	Rinse4x	CtrlGen+	UnMarker	
MS-COCO [24]													
FID ↓	90.97	45.07	112.83	64.29	110.92	44.91	42.12	42.36	47.92	51.18	40.96	49.85	40.18
CLIP ↑	0.312	0.325	0.307	0.312	0.320	0.324	0.328	0.327	0.319	0.302	0.323	0.316	0.328
SD-Prompts [34]													
FID ↓	121.34	53.29	134.29	77.32	130.60	53.21	51.99	49.78	52.06	58.83	48.95	55.48	49.47
CLIP ↑	0.338	0.352	0.330	0.342	0.346	0.352	0.354	0.358	0.356	0.331	0.364	0.353	0.364
DiffusionDB [38]													
FID ↓	107.56	53.89	129.11	78.39	121.38	52.71	48.10	47.83	51.96	55.89	47.33	50.69	47.11
CLIP ↑	0.316	0.345	0.319	0.347	0.339	0.348	0.350	0.348	0.341	0.339	0.348	0.344	0.350

Table 3. Average image-quality metrics (FID and CLIP-Text Score) across 15 watermarking schemes and three datasets. Lower FID reflects improved perceptual fidelity, while higher CLIP-Text scores indicate better semantic consistency with the prompt. Best results are highlighted in **bold**.

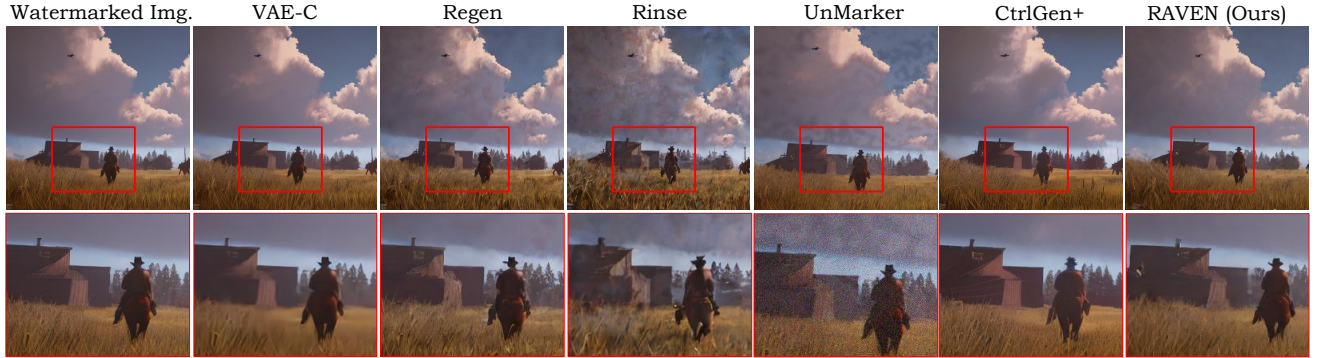


Figure 2. **Qualitative comparison of watermark removal methods.** The top row shows full images, while the bottom row displays zoomed-in regions (red boxes) for detailed inspection. VAE-C [6] introduces excessive blurring that degrades fine details. Regen [49] produces visible artifacts due to high noise injection required for watermark removal. Rinse exhibits unnatural color shifts and loss of photorealism, a consequence of performing multiple regeneration passes. UnMarker [20] leaves noisy residual artifacts that compromise visual quality. CtrlGen+ [25] produces overly stylized outputs that deviate from natural appearance. In contrast, RAVEN preserves fine-grained details, natural textures, and photorealistic appearance. Note that the zoomed regions for UnMarker [20] and RAVEN differ slightly from other methods due to cropping layers and camera translation transformations, respectively. More results are provided in Suppl.

Table 4. Image fidelity under bounded global translation on DiffusionDB dataset [38]. Evaluation is restricted to the overlapping region to account for spatial shift. Best results in **bold**.

Metric	Regen	Rinse2×	CtrlGen+	UnMarker	RAVEN
PSNR↑	25.11	24.27	21.92	21.08	25.34
SSIM↑	0.74	0.69	0.68	0.61	0.72

lation applies diagonal translations randomly sampled from [24, 32] or $[-32, -24]$ pixels along each axis. All experiments run on a single NVIDIA A100 GPU with fixed random seeds for reproducibility.

Evaluation Metrics. Following [20, 25], we report TPR@1%FPR for semantic watermarking and Bit Accuracy for bitstream-based schemes. Generation quality is assessed via FID [16] and semantic alignment via CLIP score using OpenCLIP-ViT/G [7]. We also report PSNR and SSIM under bounded global translation. Specifically, we compute these metrics over the overlapping region only, since RAVEN performs novel view synthesis that generates coherent content in newly exposed boundary regions that lack pixel-wise correspondence under any translation. This ensures a fair pixel-level comparison that accounts for the spatial shift introduced by latent viewpoint modulation.

5.2. Main Results

Watermark Removal Performance. Tab. 2 presents watermark removal results across 15 watermarking methods on MS-COCO and SD-Prompts, reporting TPR@1%FPR for semantic methods (lower is better) and Bit Accuracy for bitstream methods (0.5 indicates randomization). Classical attacks show limited effectiveness, regeneration-based methods achieve moderate suppression, and UnMarker [20] achieves the best baseline performance at 0.078 average TPR on MS-COCO. RAVEN outperforms all baselines with 0.026 average TPR on MS-COCO, representing more than 60% improvement over the closest baseline, with consistent gains across both datasets and bitstream-based schemes. Full quality-detectability trade-off curves and DiffusionDB results are provided in Suppl.

Quality Preservation. Tab. 3 reports FID and CLIP-Text scores averaged across 15 watermarking schemes on three datasets. Signal processing attacks introduce substantial distortion, with FID ranging from 45.07 to 134.29. Regeneration-based methods achieve better fidelity, with VAE-C and Regen showing FID around 42 to 52 across datasets. RAVEN achieves the best or near-best scores across all datasets: FID of 40.18, 49.47, and 47.11 on MS-COCO, SD-Prompts, and DiffusionDB respectively,



Figure 3. **Effect of strength parameter s on watermark removal.** Increasing s enhances watermark suppression but degrades visual quality. Low values ($s = 0.05$) preserve quality but may retain watermarks, while high values introduce artifacts.

Table 5. **Effect of strength parameter on visual quality.**

Strength (s)	0.10	0.15	0.20	0.25	0.35	0.45
FID ↓	62.68	65.50	69.01	75.43	79.62	85.10



Figure 4. **Effect of view-guided correspondence attention.** Without correspondence attention (middle), latent viewpoint modulation causes severe structural distortions. With view-guided attention (right), RAVEN preserves fine-grained details, textures, and structural consistency while effectively removing watermarks.

while maintaining the highest CLIP scores of 0.328, 0.364, and 0.350. Tab. 4 further reports PSNR and SSIM under bounded global translation on the DiffusionDB dataset [38], where RAVEN achieves the highest PSNR of 25.34 dB and competitive SSIM, confirming strong pixel-level fidelity in the overlapping region. These results demonstrate that RAVEN effectively balances watermark removal with superior perceptual fidelity and semantic consistency.

Qualitative Results. Fig. 2 compares the outputs of different watermark removal attacks. Pixel-space baselines struggle with quality preservation: VAE-C [6] excessively blurs details, Regen [49] introduces regeneration artifacts, and Rinse produces unnatural textures. UnMarker [20] leaves residual noise patterns, while CtrlGen+ [25] yields overly stylized, over-smoothed outputs. In contrast, RAVEN preserves fine-grained details and natural textures. Additional qualitative results are provided in Suppl.

5.3. Ablations

Effect of Strength Parameter. The strength parameter s controls noise injection during partial diffusion inversion, trading off watermark removal and visual quality. Fig. 3 and Tab. 5 show that low s preserves quality but may retain watermarks, while high s improves suppression but introduces artifacts and degrades FID.

Impact of View-Guided Correspondence Attention.

Table 6. **Model-agnostic watermark removal.** RAVEN achieves consistent watermark suppression across different Stable Diffusion Image-to-Image backbones, demonstrating generalizability without model-specific tuning.

Model	TPR@1%FPR ↓	Bit Acc.	FID ↓	CLIP ↑
Img2Img v2.1	0.026	0.523	40.18	0.328
Img2Img v2.0	0.026	0.519	42.61	0.328
Img2Img v1.5	0.029	0.547	44.93	0.325

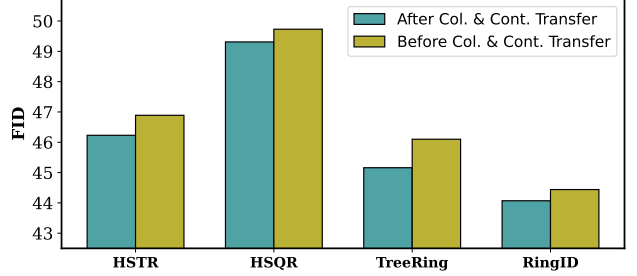


Figure 5. **Effect of color and contrast transfer.** FID comparison across four watermarking methods before and after applying color and contrast transfer in CIELAB space. The post-processing step consistently improves FID.

Fig. 4 demonstrates that removing correspondence attention causes structural distortion. Incorporating view-guided attention preserves semantic consistency and fine textures without reintroducing watermark, confirming its crucial role in maintaining perceptual quality.

Impact of Color and Contrast Transfer. Fig. 5 shows that applying color and contrast transfer in CIELAB space consistently improves FID of RAVEN outputs across all watermarking methods, correcting color shifts and contrast inconsistencies without restoring watermark signals.

Model-Agnostic Generalization. Tab. 6 shows that RAVEN achieves consistent watermark suppression across Stable Diffusion v1.5, v2.0, and v2.1 backbones without model-specific tuning. All versions maintain strong performance with TPR below 0.03 and comparable visual quality, demonstrating that our view synthesis approach generalizes across diffusion architectures.

6. Conclusion

We introduced a novel attack vector against invisible watermarks by framing removal as view synthesis. Our zero-shot diffusion-based framework applies controlled viewpoint shifts in latent space combined with view-guided correspondence attention to remove watermarks while preserving semantic content. Without requiring detector access or watermark knowledge, RAVEN achieves state-of-the-art suppression across 15 watermarking schemes and 14 attack baselines. This work exposes a critical vulnerability to semantic-preserving geometric transformations, motivating the design of more resilient watermarking mechanisms.

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